

Credit Risk

Exploratory Data Analysis — Planning Document

Version 1.0 | Data Science Division | Prepared for Internal Use

1. Project Overview & Objectives

- Dataset Goal:** Analyze credit risk data to understand the factors driving loan defaults (`loan_status`).
- Primary Objective:** Clean the dataset, explore variable distributions, uncover latent patterns, and identify relationships between borrower characteristics and loan outcomes to inform future predictive modeling.

2. Data Understanding & Dictionary

Each row in the dataset represents an individual record for a distinct borrower. Each column encodes a specific borrower attribute or loan characteristic, as defined in the table below.

Column / Variable	Description
Borrower Personal & Financial Data	
<code>person_age</code>	Age of the loan applicant, expressed in years.
<code>person_income</code>	Annual gross income of the applicant.
<code>person_home_ownership</code>	Housing status of the applicant (RENT, OWN, MORTGAGE, or OTHER).
<code>person_emp_length</code>	Total years of employment history reported by the applicant. Note: the value 123 in the first row is almost certainly a data-entry error and must be treated as an outlier.
Loan Request Data	
<code>loan_intent</code>	Stated purpose of the loan (e.g., PERSONAL, EDUCATION, MEDICAL, VENTURE, HOMEIMPROVEMENT, DEBTCONSOLIDATION).
<code>loan_grade</code>	Credit-risk classification assigned to the loan, represented by letter grades (A–E). Grade A denotes the lowest risk level; risk increases toward E.
<code>loan_amnt</code>	Total loan amount requested by the applicant.
<code>loan_int_rate</code>	Annual interest rate applied to the loan.
<code>loan_status</code>	Target Variable — Current repayment status: 0 = fully paid or compliant;

	1 = defaulted.
loan_percent_income	Loan amount expressed as a proportion of the borrower's annual income (e.g., 0.59 means the loan equals 59 % of annual income).
Credit History Data	
cb_person_default_on_file	Indicates whether the borrower has a historical record of default or financial delinquency (Y = Yes, N = No).
cb_person_cred_hist_length	Duration of the borrower's credit history in years, measured from their first interaction with a lender or credit facility.

3. Key EDA Questions & Hypotheses

The following questions frame the analytical scope of the EDA phase and drive visualization and feature-selection decisions.

#	Research Question / Hypothesis
Q1	What is the baseline default rate in the dataset? (Target-class imbalance check.)
Q2	Does a lower annual income (person_income) significantly increase the likelihood of default?
Q3	How does the interest rate (loan_int_rate) affect a borrower's ability to repay?
Q4	Are specific loan intents (e.g., Medical vs. Venture) more strongly associated with defaults?
Q5	Is there a strong correlation between a borrower's historical default record (cb_person_default_on_file) and current loan status?

4. Data Cleaning & Preprocessing Plan

Initial data inspection revealed anomalies that must be addressed prior to in-depth visualization. The three focus areas below define the preprocessing actions required.

Critical Notice — Resolve all items below before proceeding to the visualization phase.

Outliers — person_emp_length contains physically impossible values (e.g., 123 years). Inspect the full distribution and apply capping or removal as appropriate.

Outliers — Examine person_age for unrealistic values (e.g., ages below 18 or above 100).

Missing Values — Scan loan_int_rate and person_emp_length for null or missing values. Define an imputation strategy (mean or median) based on each variable's distribution and skewness.

5. Visualization Strategy

The table below serves as the chart blueprint for the EDA phase. Each row maps an analysis type to the appropriate visualization and articulates its analytical objective.

Analysis Type	Chart Type	Description & Goal
Target Distribution	Bar / Pie Chart	Visualize the ratio of Default (1) vs. Non-Default (0) cases to assess class imbalance.
Univariate — Numerical	Histograms	Plot person_income, loan_amnt, and person_age to assess distribution shape and skewness.
Univariate — Categorical	Bar Charts	Display loan counts segmented by loan_intent, loan_grade, and person_home_ownership.
Multivariate Relationships	Correlation Heatmap	Construct a correlation matrix of all numerical features to identify multicollinearity (e.g., Age vs. Credit History Length).
Feature vs. Target (Numerical)	Box Plots	Compare distributions of loan_int_rate and person_income across loan_status groups.
Feature vs. Target (Categorical)	Stacked Bar Charts	Show the default rate within each loan_grade and person_home_ownership category.